

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Concept Mapping Questions

Course Code:	CSA0303	Course Title:	Data Structures
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Mapping Questions
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:	N.Thejasree	Reg. No.:	192524438
Section / Batch:	Sloat c	Date:	

Code of Conduct

I N.Thejasree(192524438) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.Thejasree

Instructions

- This test contains 3 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C'	Concept mapping on classifications of data structures.	1	30
Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.	Linear and Binary search	2	40
Primitive and non-primitive data structure, Linear and non-linear data structure Arrays	Linked List	3	40
GRAND TOTAL:			100 Marks

CO2 - DP1 Analytical.

N. Thyasree

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AIDS

CSA0303 - Data structures

26
35

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perform the following operation in LIFO data structure having an array size of 5

Push (45)

Push (22)

Push (11)

Push (77)

Pop ()

Push (44)

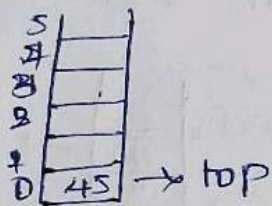
Push (33)

Pop ()

Pop ()

Pop ()

1. Push (45) Size = 5



$top = top + 1$

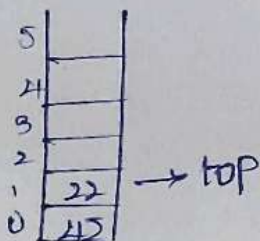
$top = 0 + 1$

$top = 1$

if it is Empty

$top = -1$

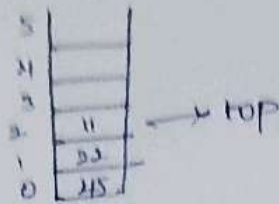
Push (22)



$top = 1 + 1$

$top = 2$

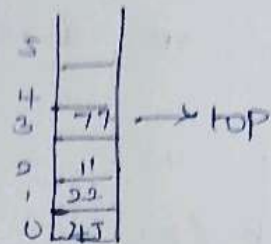
III. push(11)



$$\text{top} = 2 + 1$$

$$\text{top} = 3$$

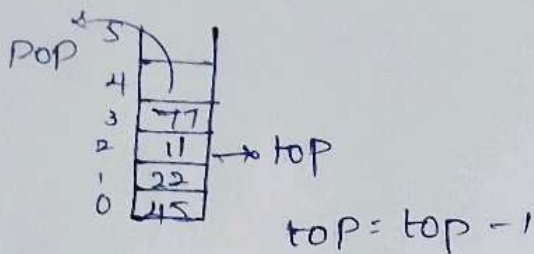
IV. push(77)



$$\text{top} = 3 + 1$$

$$\text{top} = 4$$

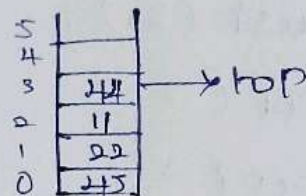
V. pop()



$$\text{top} = 3 - 1$$

$$\text{top} = 2$$

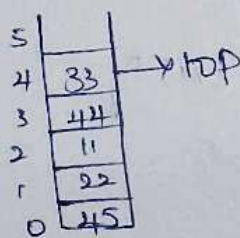
VI. push(44)



$$\text{top} = 3 + 1$$

$$\text{top} = 4$$

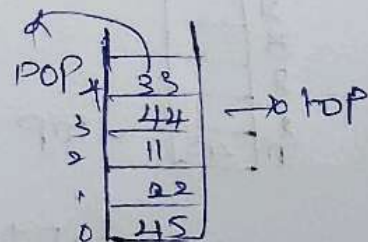
VII. push(33)



$$\text{top} = 4 + 1$$

$$\text{top} = 5$$

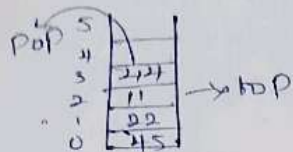
VIII. pop()



$$\text{top} = 4 - 1$$

$$\text{top} = 3$$

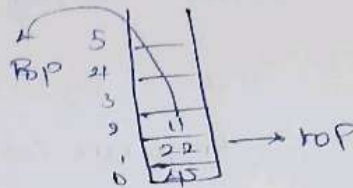
x. popc)



$$\text{top} = 3 - 1$$

$$\text{top} = 2$$

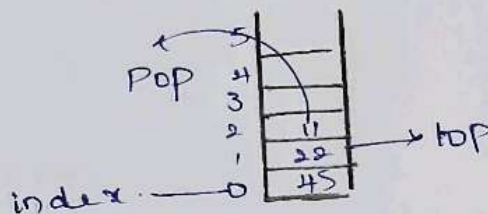
x. popc)



$$\text{top} = 2 - 1$$

$$\text{top} = 1$$

∴ Peak element = 22
index position = 1



10

perform the following operations in Circular Queue, having a size of 5.

Size = 5

Enqueue (20)

Enqueue (10)

Enqueue (30)

Dequeue ()

Enqueue (80)

Dequeue (),

Enqueue (90)

Enqueue (100)

Dequeue (),

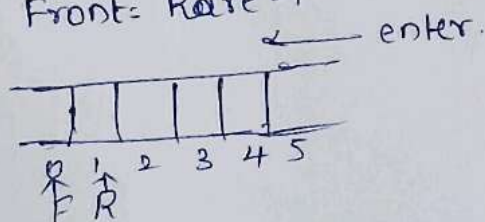
Dequeue (60),

Enqueue (90).

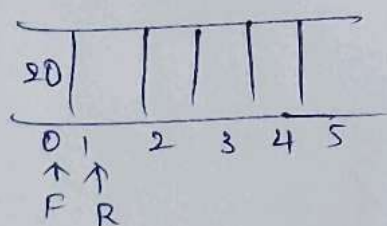
Size = 5

Front = Rear - 1

Rear = Front + 1

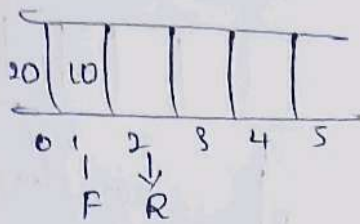


(i) Enqueue (20)



Front = Rear - 1
= 1 - 1 = 0.

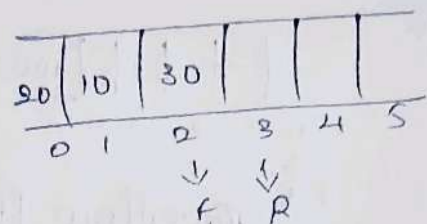
(i) Enqueue (10)



$$\text{Front} = 2 - 1$$

$$\text{Front} = 1$$

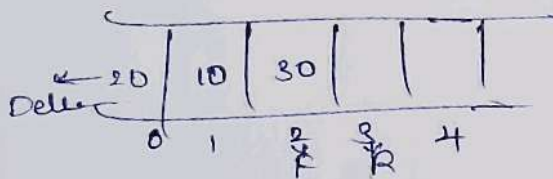
(iii) Enqueue (30)



$$\text{Front} = 3 - 1$$

$$\text{Front} = 2$$

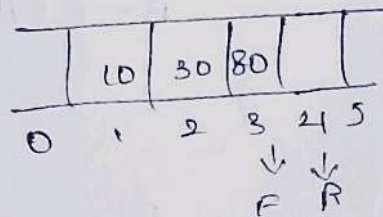
(iv) Dequeue ()



$$\text{Front} = 3 - 1$$

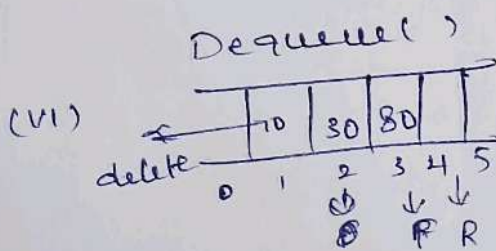
$$\text{Front} = 2$$

(v) Enqueue (80)



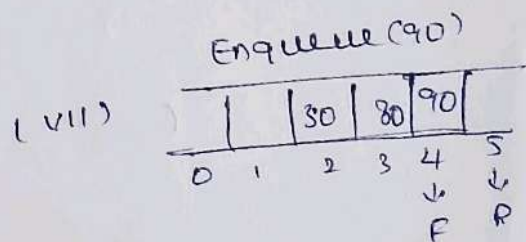
$$\text{Front} = 4 - 1$$

$$\text{Front} = 3$$



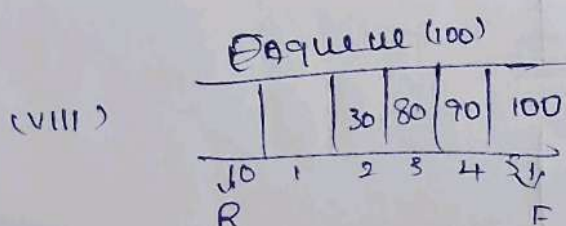
$$\text{Front} = 4 - 1$$

$$\text{Front} = 3$$



$$\text{Front} = 5 - 1$$

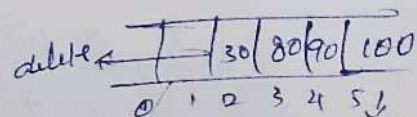
$$\text{Front} = 4$$



$$\text{Front} = 0 - 1$$

$$= 5$$

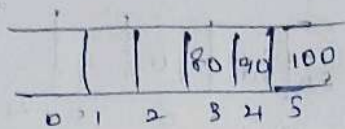
(ix) Dequeue ()



$$\text{Front} = 0 - 1$$

$$= 5$$

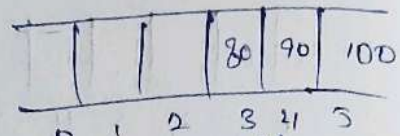
(xii) Dequeue (60)



FR
overflow $\boxed{\text{top} = \text{N} - 1}$

The queue is full

(xiii) Dequeue (90)

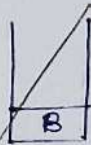


FR
overflow.

3. Convert the following infix expression into a post expression.

$$B - (C/D + (E \times F * G) / H) * J$$

B



B



B *

C



C



/



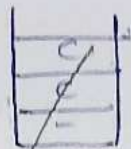
D



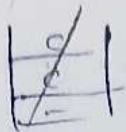
+



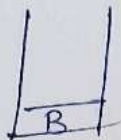
C



E



B



B

-



B

C



B

C



BC

/



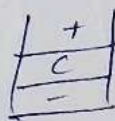
BC

D



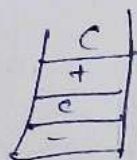
BCD

+



BCD /

C



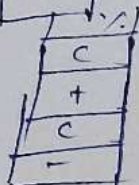
BCD /

E

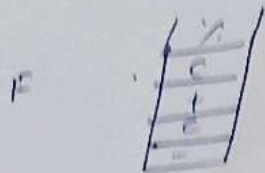


BCD/E

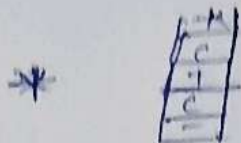
%



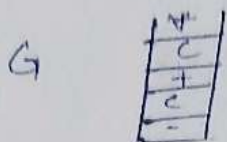
BCD/E.



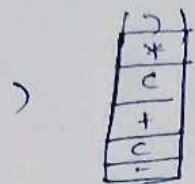
BCD/EF



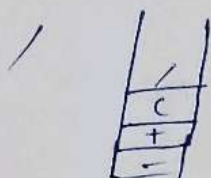
BCD/EF%



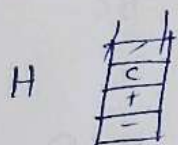
BCD/EF%G



BCD/EF%G-



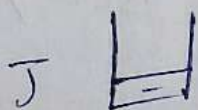
BCD/EF%G.*



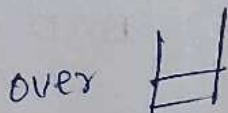
BCD/EF%G*H



BCD/EF%G*H/+



BCD/EF%G*H/+J



BCD/EF%G*H/+J*

(error)

(0) 9 3 8 4 / * +

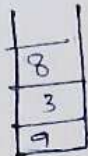
9



3



8



4



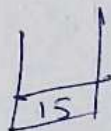
/



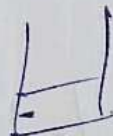
*



+



over



$$9/8 = 2$$

$$3 * 2 = 6$$

$$9 + 6 = 15$$

$$0/P = 15$$



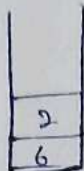
(b)

$$6 \div 6 = 8 \div *$$

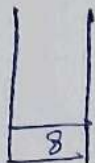
6



2



+



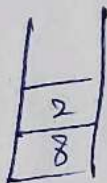
8



3



/

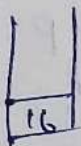


*



8

over



$$6 + 2 = 8$$

$$3 / 6 = 2$$

$$8 * 2 = 16$$

$$0 / P = 16$$

A – Concept mapping questions

- 1) A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.
- 2) A system needs to search for specific records in a large dataset efficiently. Construct a concept map linking Searching Algorithm → Linear Search → Binary Search → Time Complexity → Efficiency. Include relationships that compare the number of operations and performance of each method. Analyze which algorithm is more efficient and justify your conclusion.
- 3) A library management system stores thousands of book records including title, author, and availability status. Construct a concept map linking Data → Data Structure → Organization → Access Methods → Efficiency. Include different data structures used for storing and retrieving records. Explain how proper organization improves search efficiency and system performance.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____